

AKADEMIA GÓRNICZO-HUTNICZA
IM. STANISŁAWA STASZICA W KRAKOWIE

AGH UNIVERSITY OF SCIENCE
AND TECHNOLOGY

Electrostatic Structural Analysis in Marc Mentat

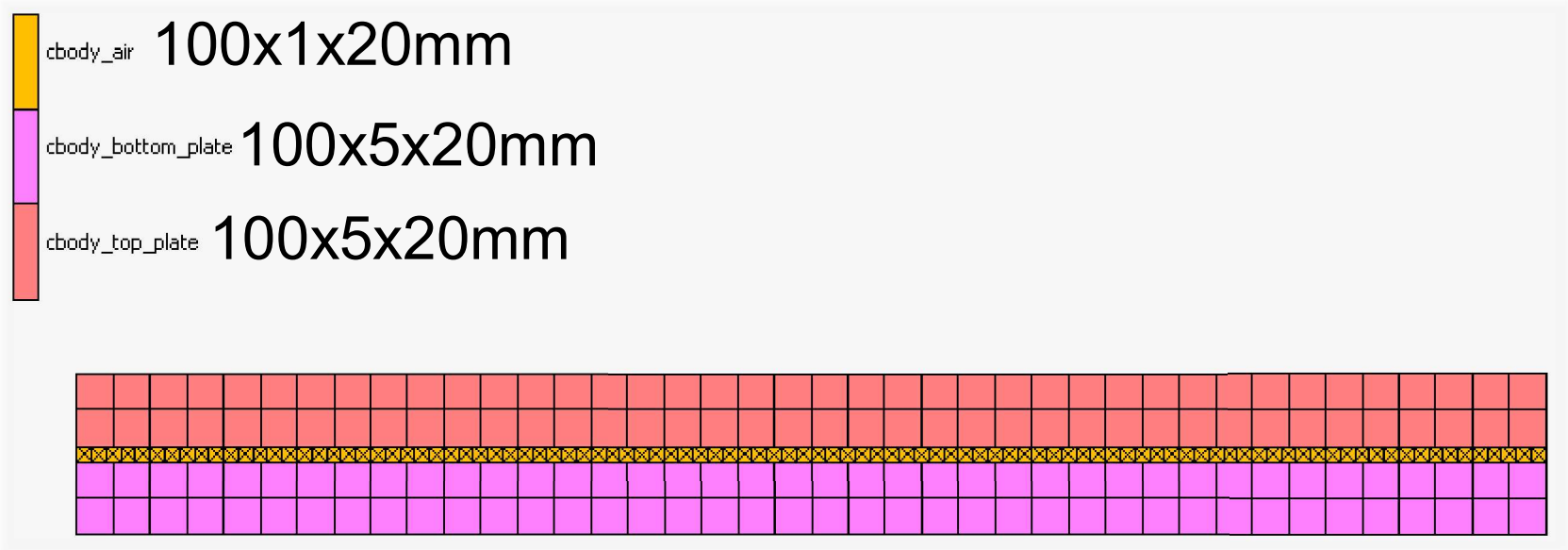
Created by Jakub Bryła

Agenda:

1. Geometry
2. Material Properties
3. Contact
4. Boundary condition
5. Loadcases
6. Job
7. Run job and save results
8. Final task

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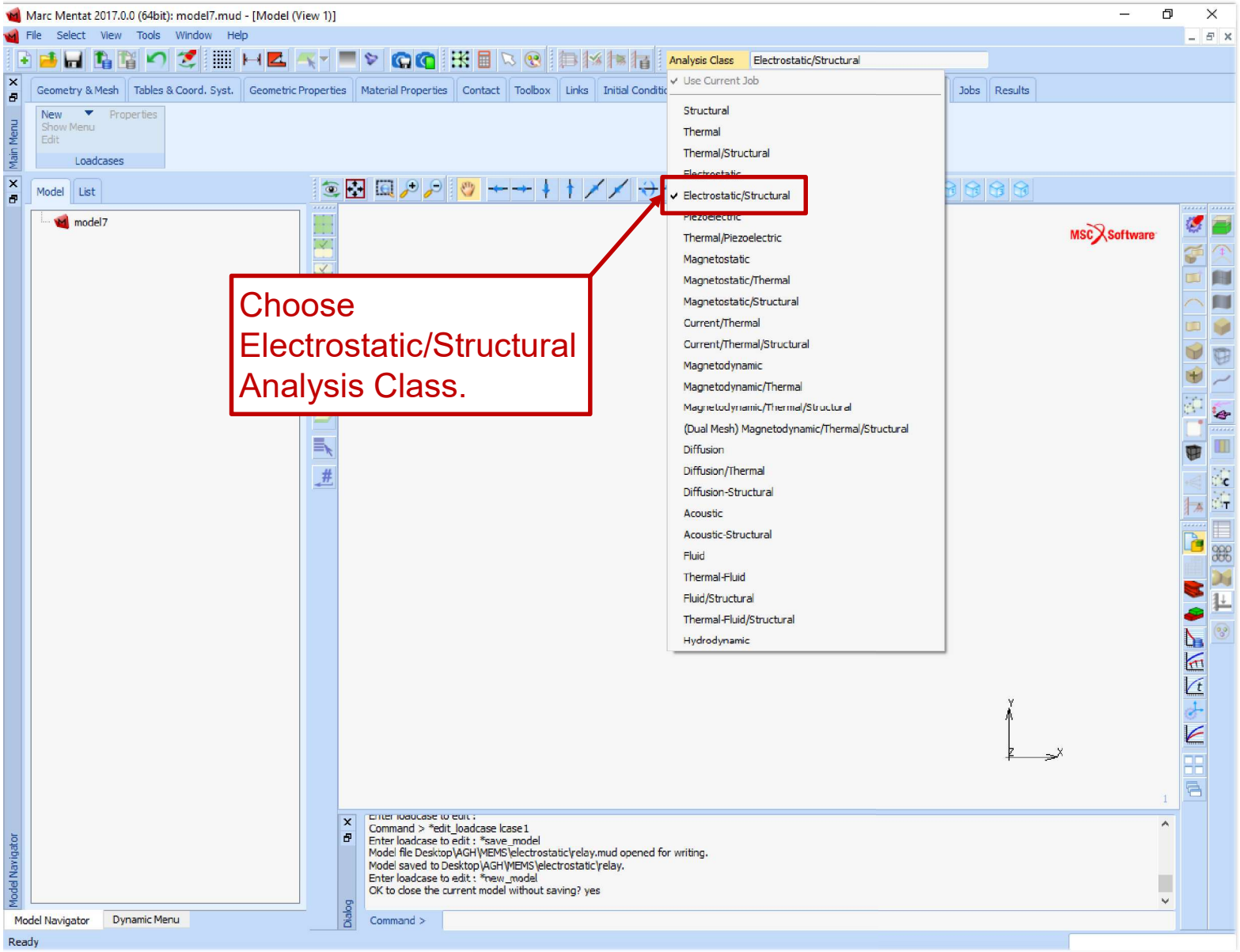
Geometry



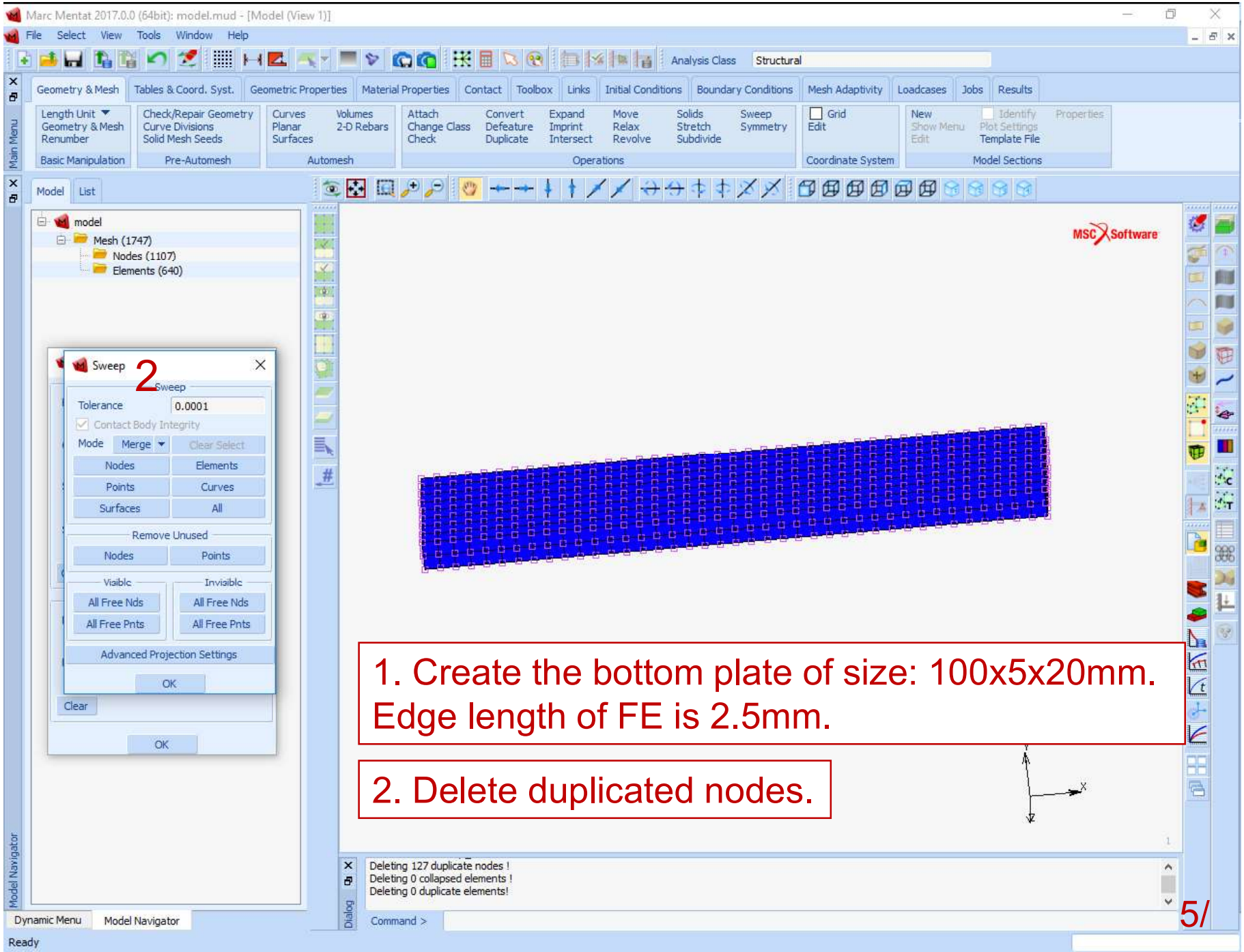
Note: plates and air block have separate nodes in the same location.

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Geometry

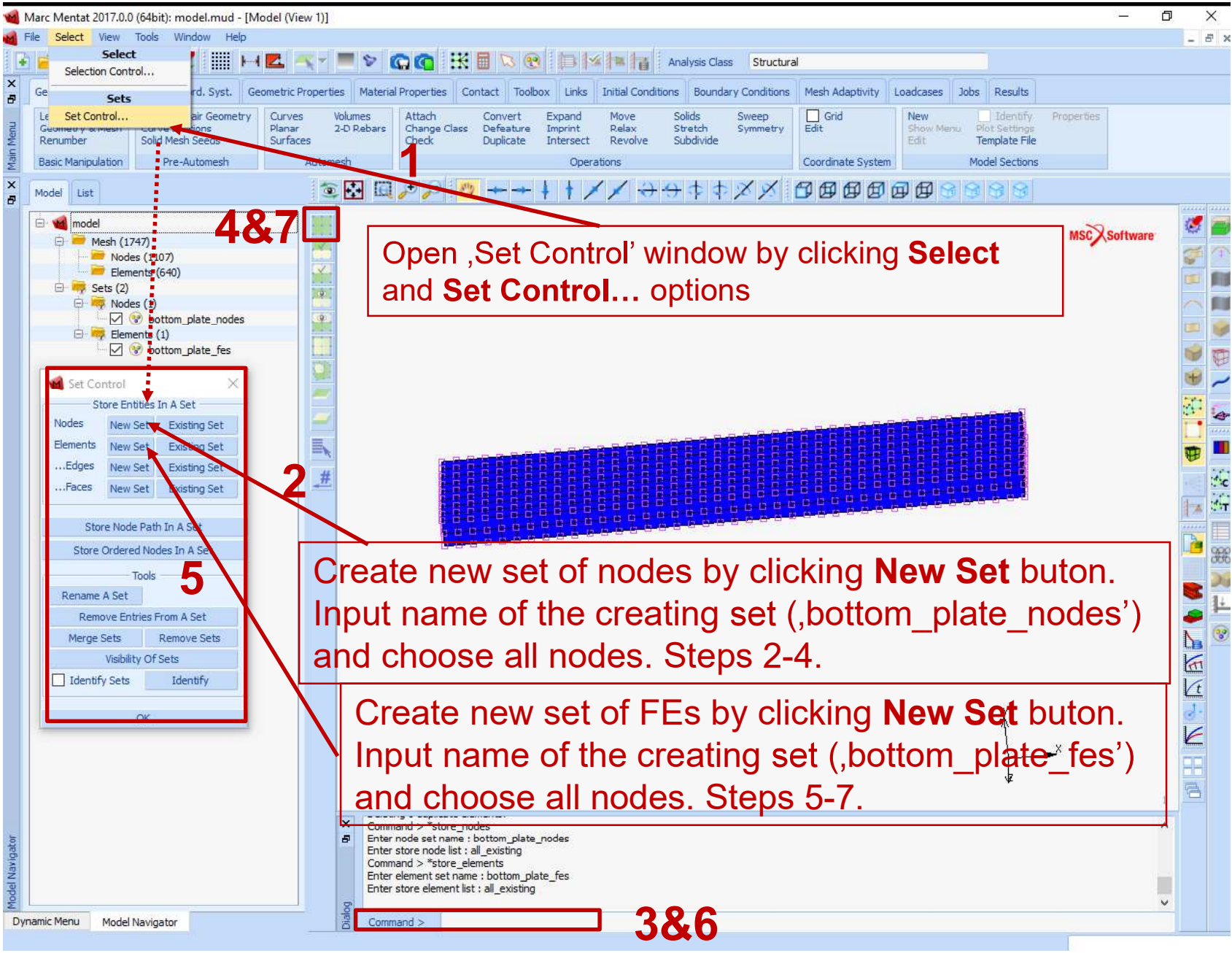

Marc Mentat 2017.0.0 (64bit): model7.mud - [Model (View 1)]
File Select View Tools Window Help
Analysis Class
Electrostatic/Structural
Use Current Job
Structural
Thermal
Thermal/Structural
Electrostatic/Structural
Thermal/Piezoelectric
Magnetostatic
Magnetostatic/Thermal
Magnetostatic/Structural
Current/Thermal
Current/Thermal/Structural
Magnetodynamic
Magnetodynamic/Thermal
(Dual Mesh) Magnetodynamic/Thermal/Structural
Diffusion
Diffusion/Thermal
Diffusion-Structural
Acoustic
Acoustic-Structural
Fluid
Thermal-Fluid
Fluid/Structural
Thermal-Fluid/Structural
Hydrodynamic
Geometry & Mesh
Tables & Coord. Syst.
Geometric Properties
Material Properties
Contact
Toolbox
Links
Jobs
Results
Loadcases
Model
List
model7
Choose Electrostatic/Structural Analysis Class.
Command > *edit_loadcase lcase 1
Enter loadcase to edit : *save_model
Model file Desktop\AGH\MEMS\electrostatic\relay.mud opened for writing.
Model saved to Desktop\AGH\MEMS\electrostatic\relay.
Enter loadcase to edit : *new_model
OK to close the current model without saving? yes
Model Navigator
Dynamic Menu
Command >
Ready

Geometry

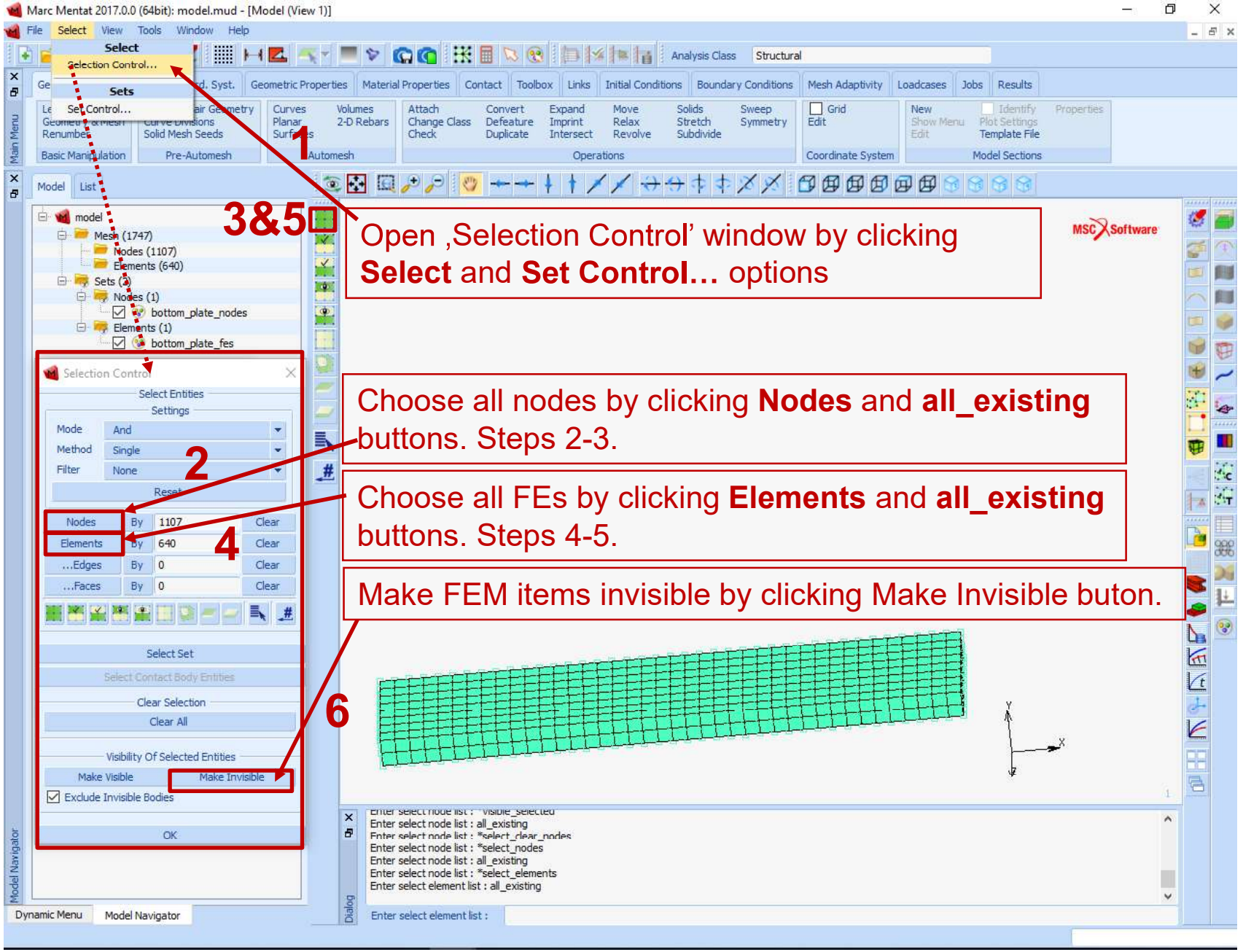


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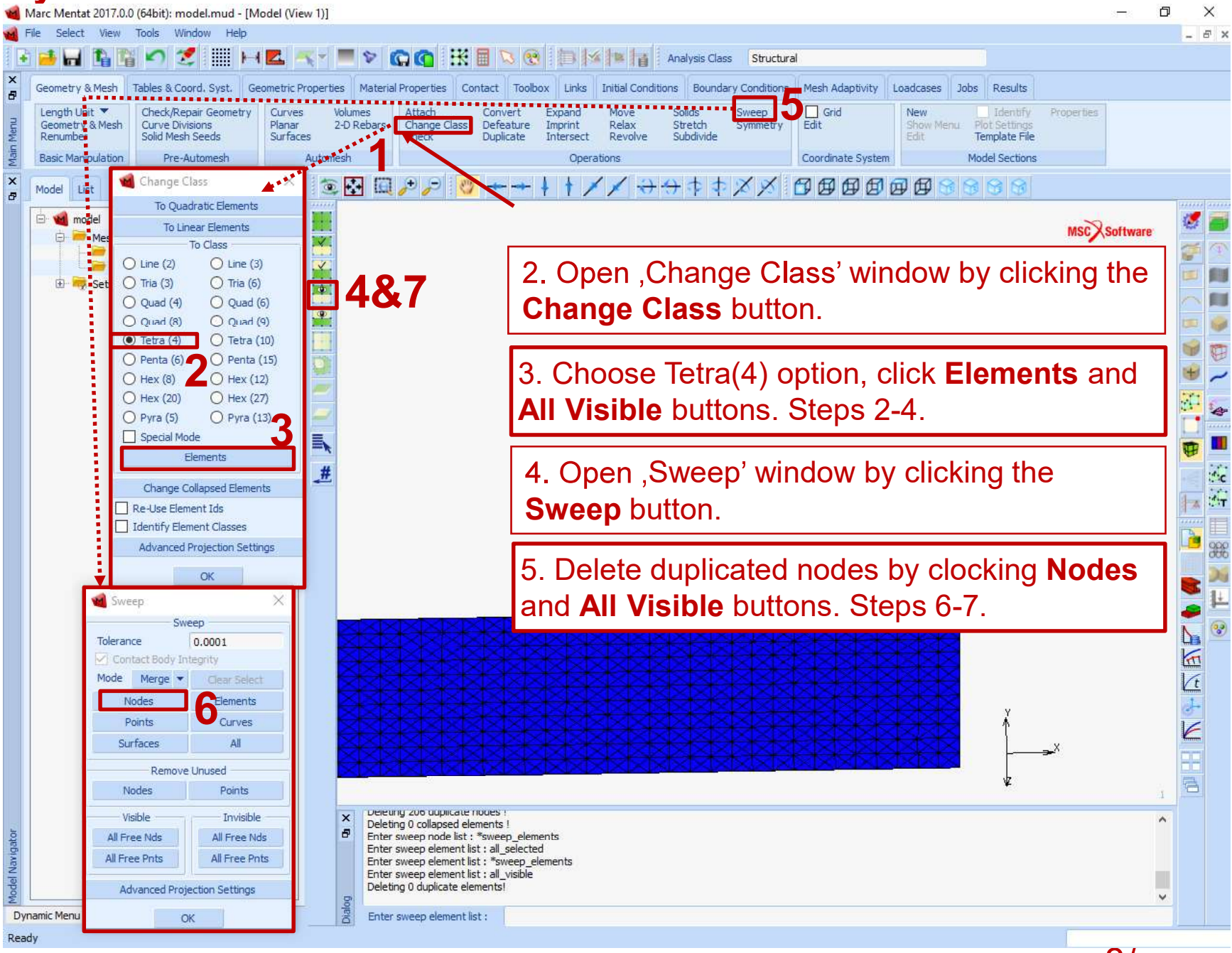
Geometry


Marc Mentat 2017.0.0 (64bit): model.mud - [Model (View 1)]
File Select View Tools Window Help
Select
Selection Control...
Sets
Set Control...
Analysis Class
Structural
Geometric Properties
Material Properties
Contact
Toolbox
Links
Initial Conditions
Boundary Conditions
Mesh Adaptivity
Loadcases
Jobs
Results
Renumber
Basic Manipulation
Curves
Planar
Surfaces
Volumes
2-D Rebars
Attach
Change Class
Check
Convert
Defeature
Duplicate
Expand
Imprint
Intersect
Move
Relax
Revolve
Solids
Stretch
Subdivide
Sweep
Symmetry
Operations
Grid
Edit
Coordinate System
New
Show Menu
Edit
Identify
Plot Settings
Template File
Properties
Model Sections
Model
List
model
Mesh (1747)
Nodes (1107)
Elements (640)
Sets (2)
Nodes (1)
bottom_plate_nodes
Elements (1)
bottom_plate_fes
Set Control
Store Entities In A Set
Nodes
Elements
...Edges
...Faces
New Set
Existing Set
Store Node Path In A Set
Store Ordered Nodes In A Set
Tools
Rename A Set
Remove Entries From A Set
Merge Sets
Remove Sets
Visibility Of Sets
Identify Sets
Identify
OK
1
2
4&7
5
Open ‚Set Control' window by clicking Select and Set Control... options
Create new set of nodes by clicking New Set buton. Input name of the creating set (‚bottom_plate_nodes') and choose all nodes. Steps 2-4.
Create new set of FEs by clicking New Set buton. Input name of the creating set (‚bottom_plate_fes') and choose all nodes. Steps 5-7.
Enter node set name : bottom_plate_nodes
Enter store node list : all_existing
Command > *store_elements
Enter element set name : bottom_plate_fes
Enter store element list : all_existing
Command >
3&6
Dynamic Menu
Model Navigator
MSC Software

Geometry


Marc Mentat 2017.0.0 (64bit): model.mud - [Model (View 1)]
Select
Selection Control...
Sets
Set Control...
1
3&5
2
4
6
Open ‚Selection Control' window by clicking **Select** and **Set Control...** options
Choose all nodes by clicking **Nodes** and **all_existing** buttons. Steps 2-3.
Choose all FEs by clicking **Elements** and **all_existing** buttons. Steps 4-5.
Make FEM items invisible by clicking Make Invisible buton.
Selection Control
Select Entities
Settings
Mode And
Method Single
Filter None
Reset
Nodes By 1107 Clear
Elements By 640 Clear
...Edges By 0 Clear
...Faces By 0 Clear
Select Set
Select Contact Body Entities
Clear Selection
Clear All
Visibility Of Selected Entities
Make Visible
Make Invisible
Exclude Invisible Bodies
OK
Enter select node list : all_existing
Enter select node list : *select_nodes
Enter select node list : all_existing
Enter select node list : *select_elements
Enter select element list : all_existing
Enter select element list :

Geometry



2. Open ‚Change Class' window by clicking the **Change Class** button.

3. Choose Tetra(4) option, click **Elements** and **All Visible** buttons. Steps 2-4.

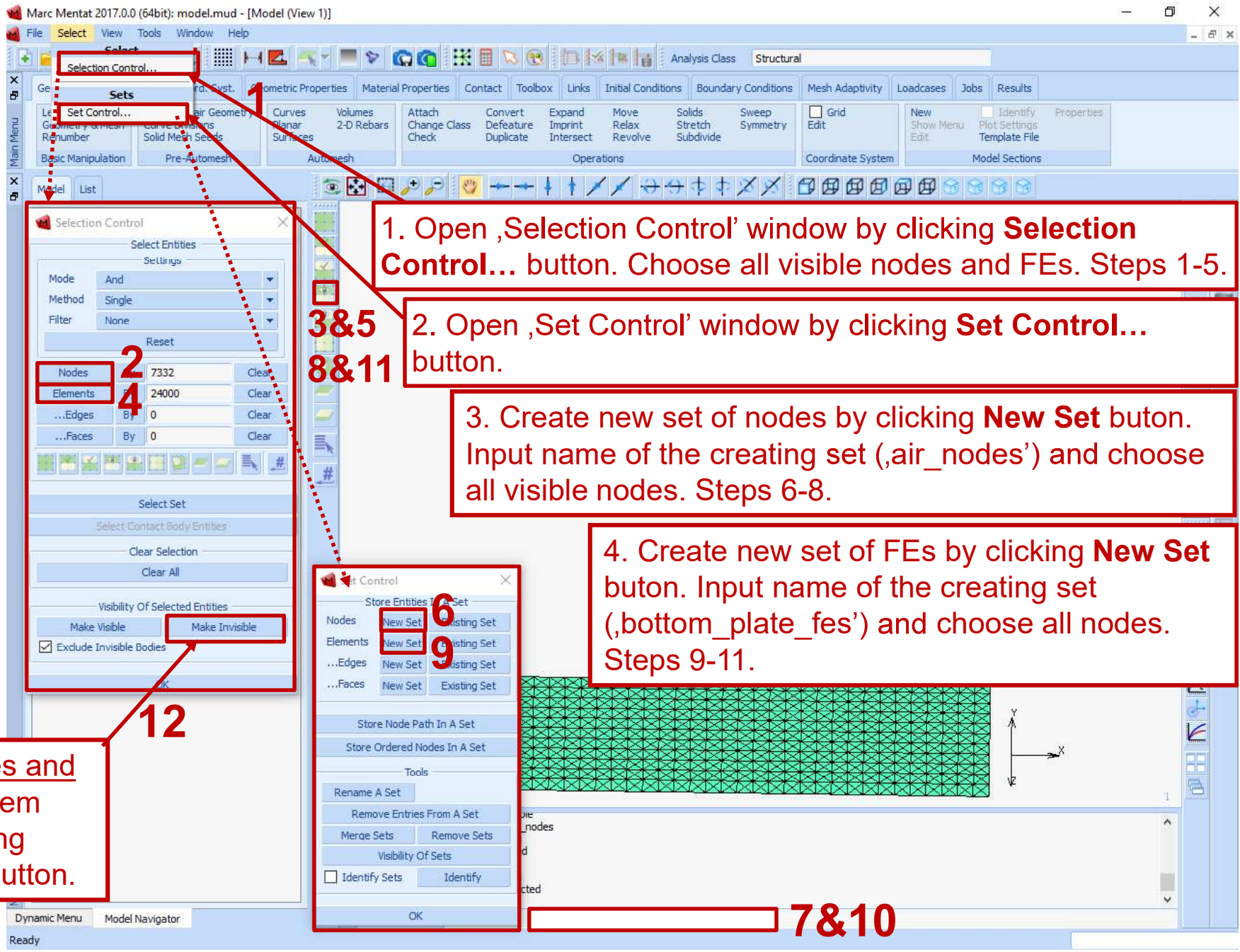
4. Open ‚Sweep' window by clicking the **Sweep** button.

5. Delete duplicated nodes by clocking **Nodes** and **All Visible** buttons. Steps 6-7.

1. Create the air block, above bottom plate, of size: 100x1x20mm.

Edge length of FE is 1mm.

Geometry



1. Open ‚Selection Control' window by clicking **Selection Control...** button. Choose all visible nodes and FEs. Steps 1-5.

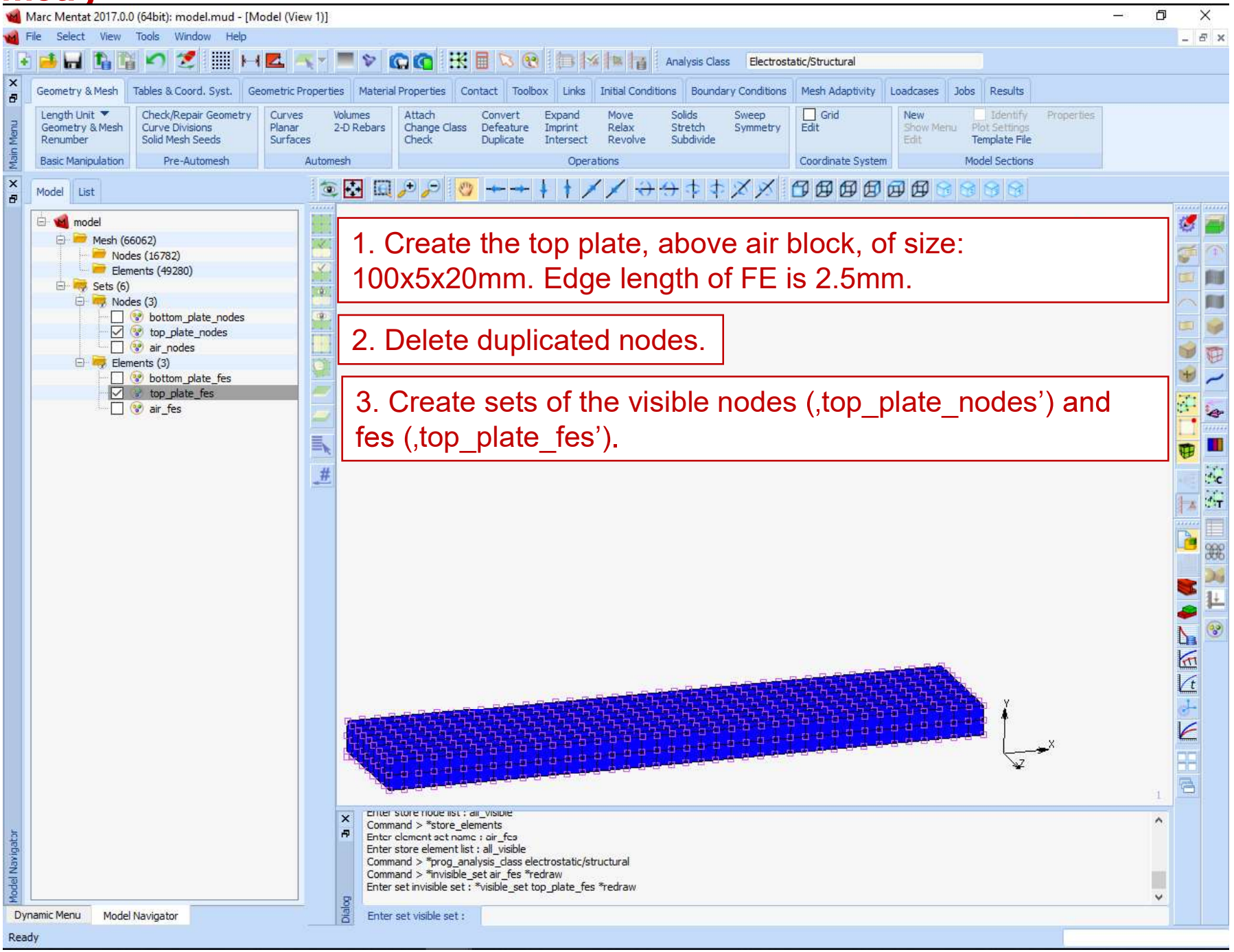
2. Open ‚Set Control' window by clicking **Set Control...** button.

3. Create new set of nodes by clicking **New Set** buton. Input name of the creating set (‚air_nodes') and choose all visible nodes. Steps 6-8.

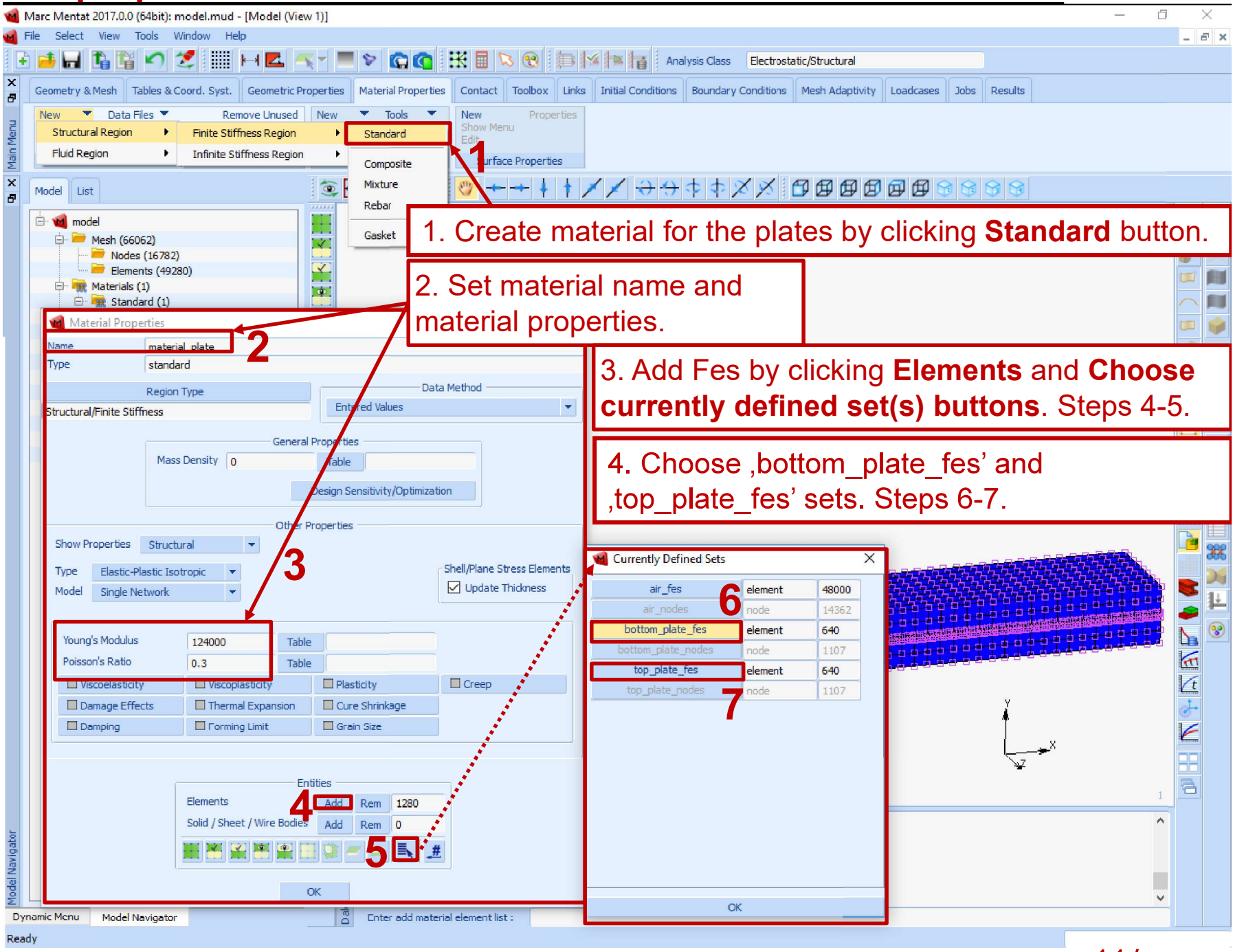
4. Create new set of FEs by clicking **New Set** buton. Input name of the creating set (‚bottom_plate_fes') and choose all nodes. Steps 9-11.

5. Select all nodes and FEs and make them invisible by clicking **Make Invisible** button.

Geometry

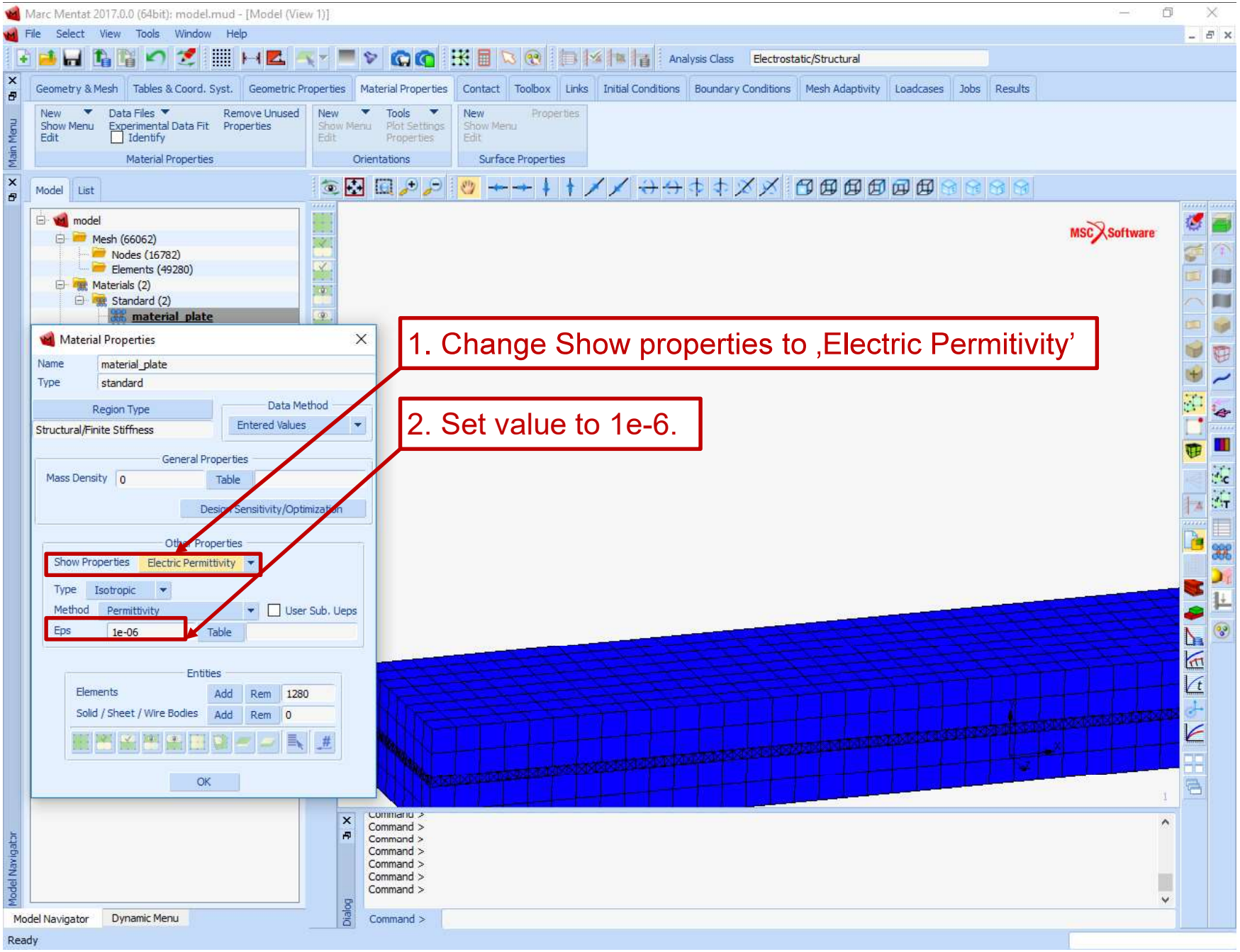

Marc Mentat 2017.0.0 (64bit): model.mud - [Model (View 1)]
File Select View Tools Window Help
Analysis Class
Electrostatic/Structural
Geometry & Mesh
Tables & Coord. Syst.
Geometric Properties
Material Properties
Contact
Toolbox
Links
Initial Conditions
Boundary Conditions
Mesh Adaptivity
Loadcases
Jobs
Results
Model
List
model
Mesh (66062)
Nodes (16782)
Elements (49280)
Sets (6)
Nodes (3)
bottom_plate_nodes
top_plate_nodes
air_nodes
Elements (3)
bottom_plate_fes
top_plate_fes
air_fes
1. Create the top plate, above air block, of size: 100x5x20mm. Edge length of FE is 2.5mm.
2. Delete duplicated nodes.
3. Create sets of the visible nodes (‚top_plate_nodes') and fes (‚top_plate_fes').
Command > *store_elements
Enter store element list : all_visible
Command > *prog_analysis_class electrostatic/structural
Command > *invisible_set air_fes *redraw
Enter set invisible set : *visible_set top_plate_fes *redraw
Enter set visible set :
Dynamic Menu
Model Navigator
Ready

Material properties



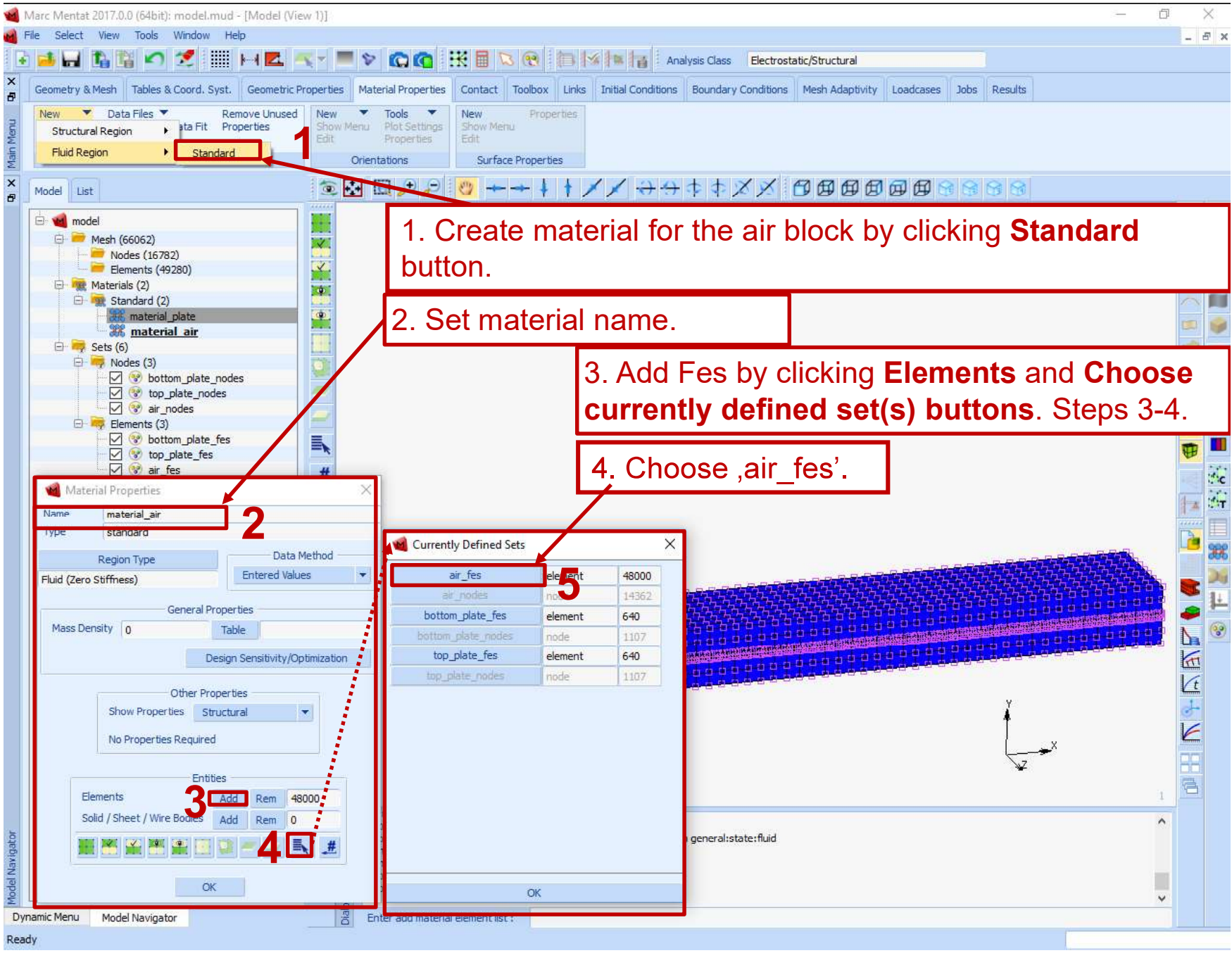
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Material properties

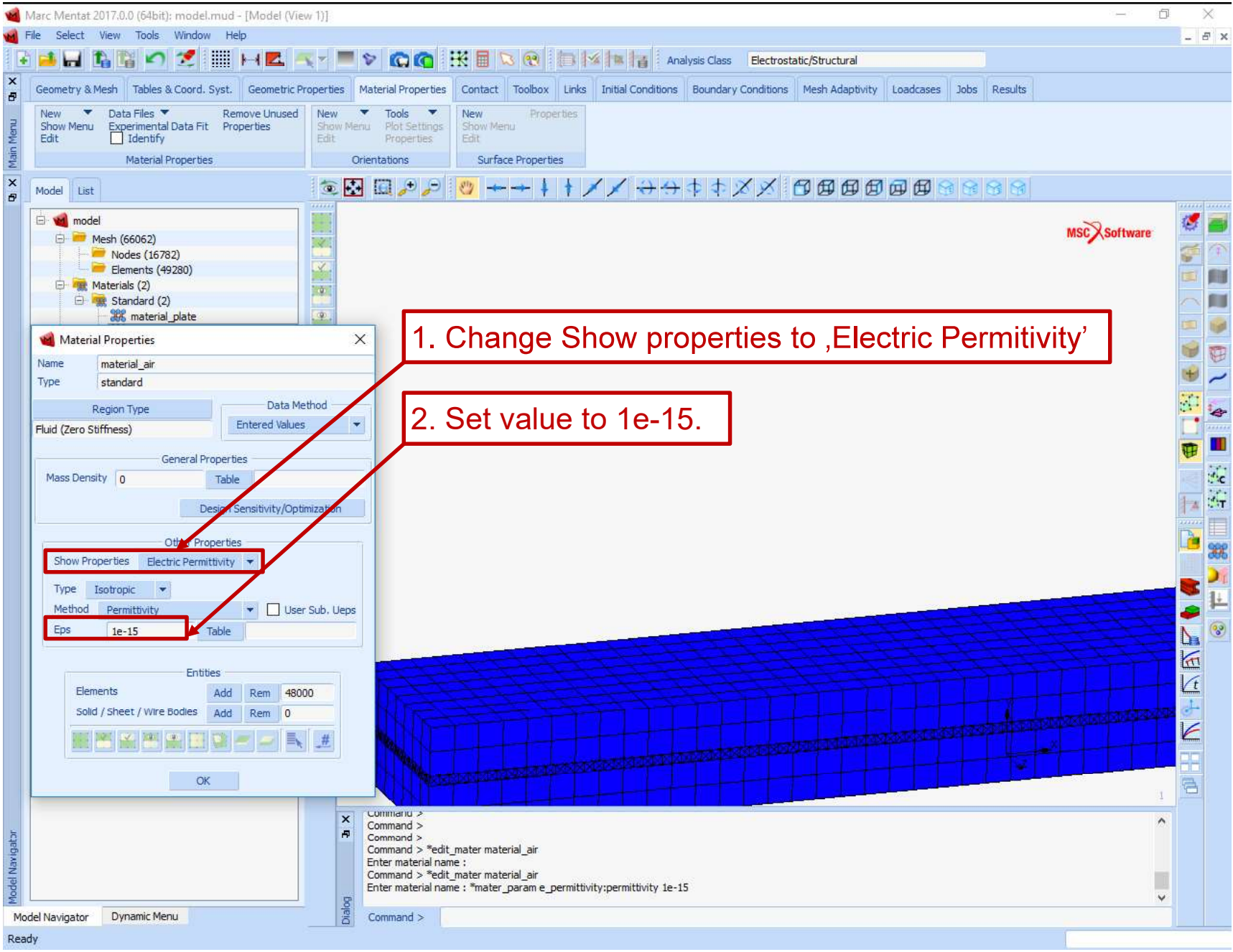


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Material properties



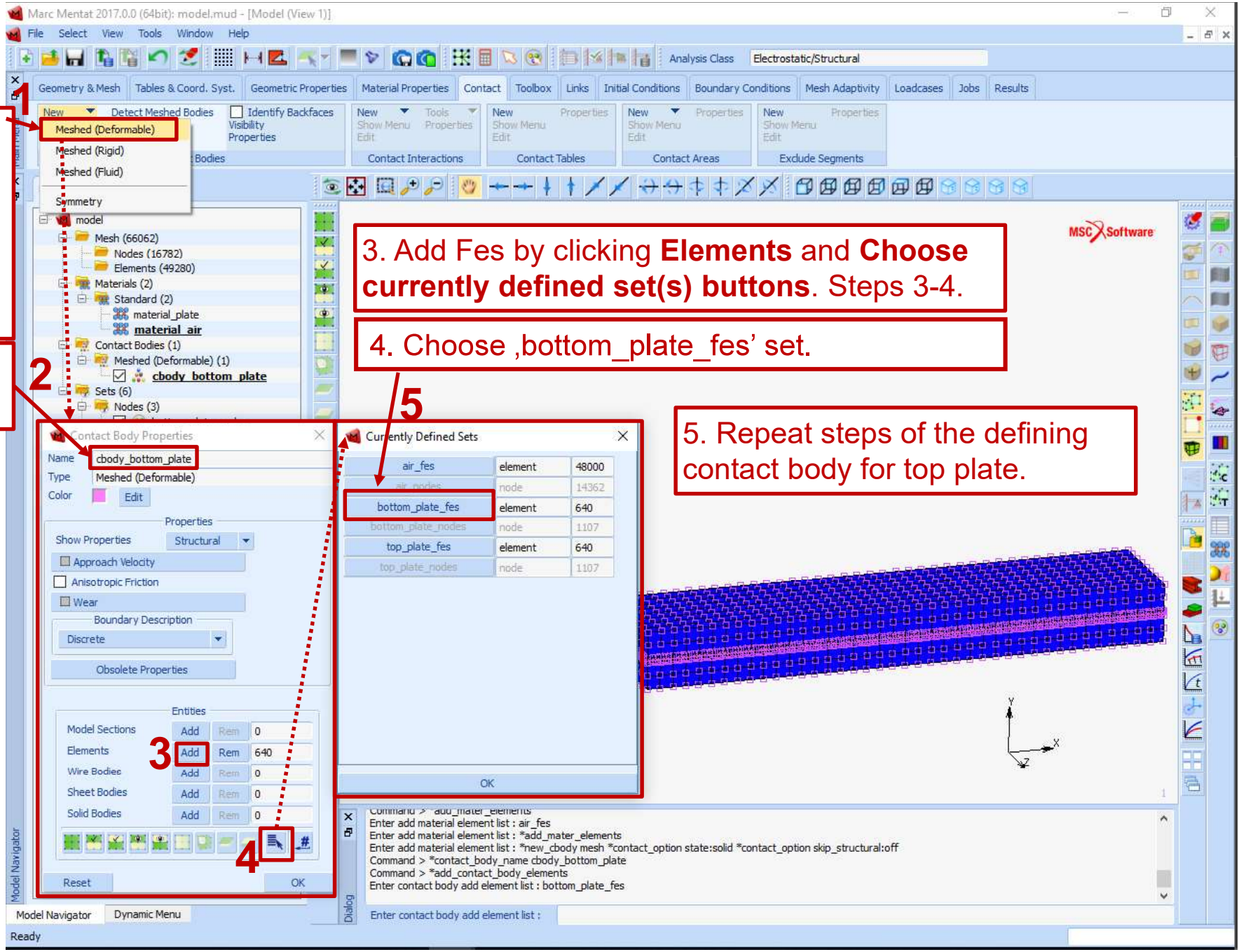
Material properties



Contact

1. Define contact body for bottom plate by clicking **Meshed (deformable)** button.

2. Set name of contact body.



3. Add Fes by clicking **Elements** and **Choose currently defined set(s) buttons**. Steps 3-4.

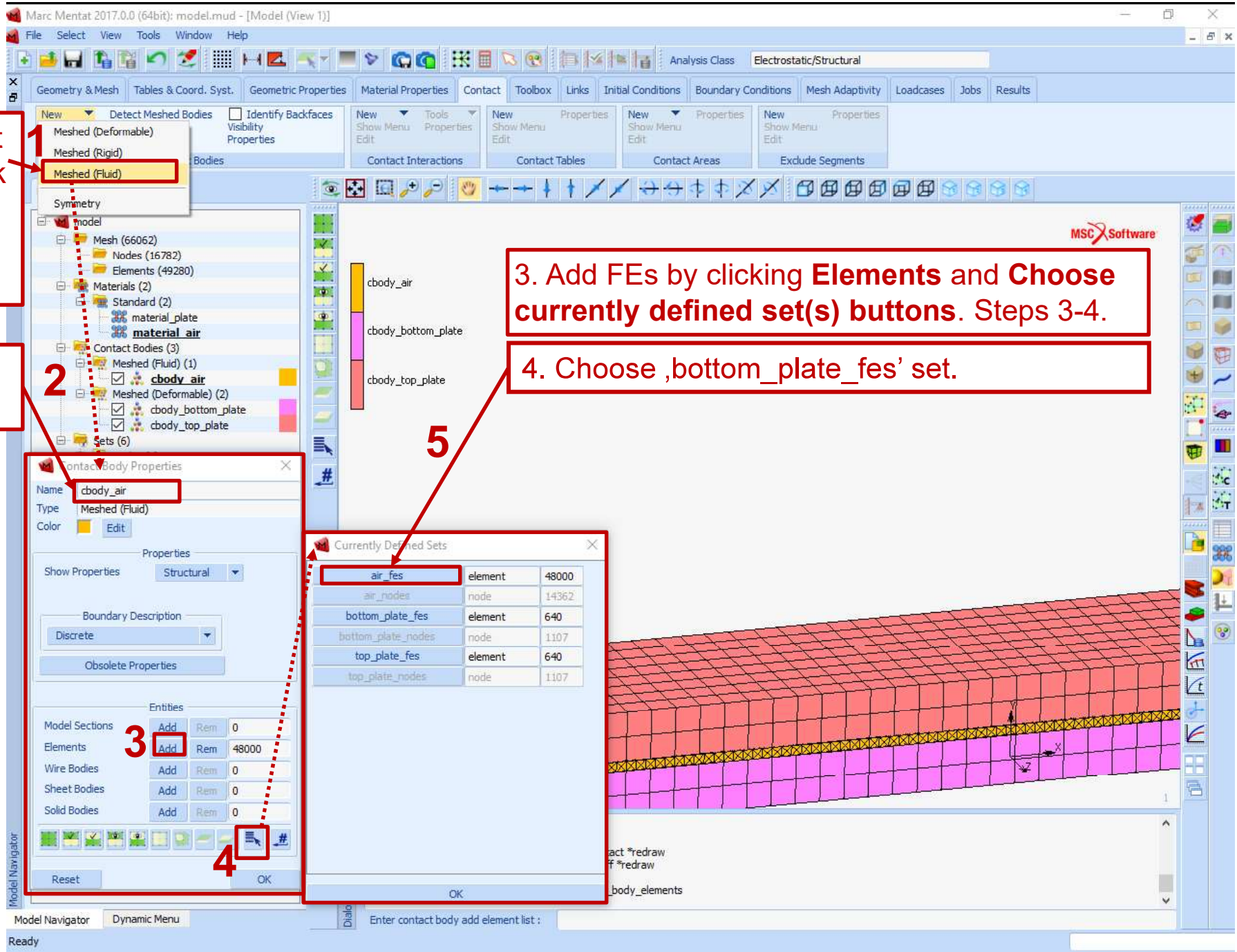
4. Choose ‚bottom_plate_fes' set.

5. Repeat steps of the defining contact body for top plate.

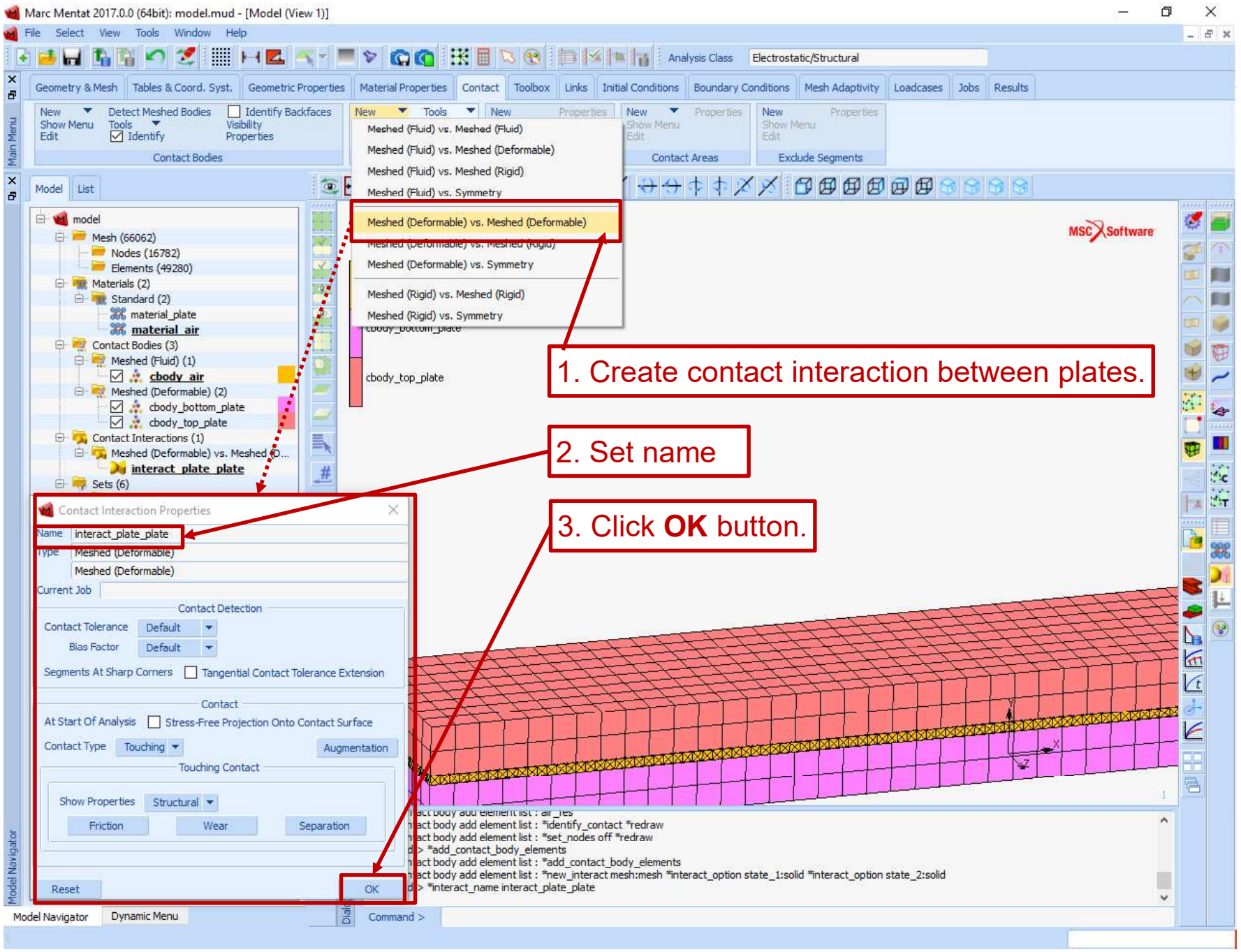
Contact

1. Define contact body for air block by clicking **Meshed (Fluid)** button.

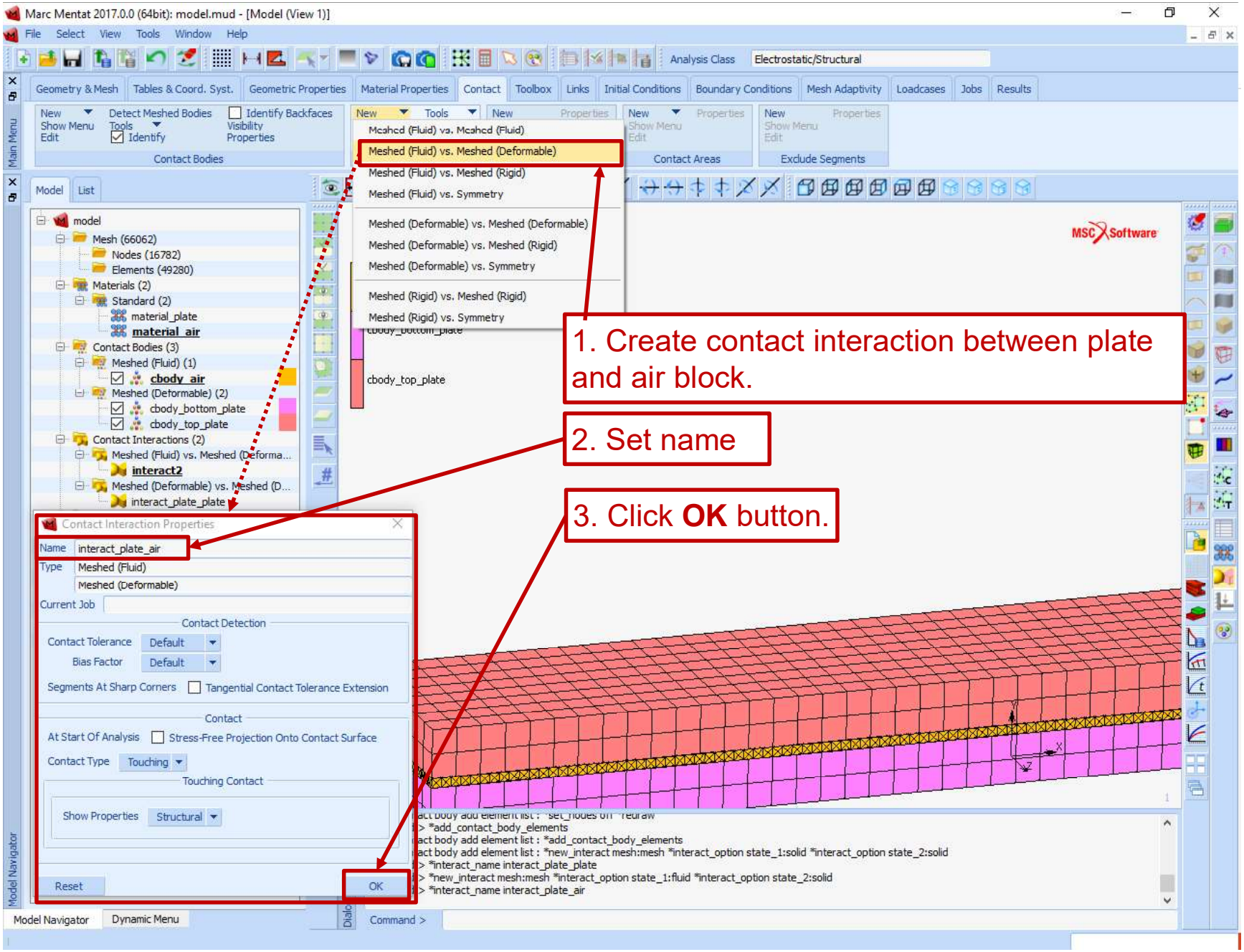
2. Set name of contact body.



Contact


Marc Mentat 2017.0.0 (64bit): model.mud - [Model (View 1)]
Meshed (Fluid) vs. Meshed (Fluid)
Meshed (Fluid) vs. Meshed (Deformable)
Meshed (Fluid) vs. Meshed (Rigid)
Meshed (Fluid) vs. Symmetry
Meshed (Deformable) vs. Meshed (Deformable)
Meshed (Deformable) vs. Symmetry
Meshed (Rigid) vs. Meshed (Rigid)
Meshed (Rigid) vs. Symmetry
1. Create contact interaction between plates.
2. Set name
3. Click OK button.
Contact Interaction Properties
Name interact_plate_plate
Type Meshed (Deformable)
Meshed (Deformable)
Current Job
Contact Detection
Contact Tolerance Default
Bias Factor Default
Segments At Sharp Corners
Tangential Contact Tolerance Extension
Contact
At Start Of Analysis
Stress-Free Projection Onto Contact Surface
Contact Type Touching
Augmentation
Touching Contact
Show Properties Structural
Friction
Wear
Separation
Reset
OK
cbody_top_plate
MSC Software

Contact


Marc Mentat 2017.0.0 (64bit): model.mud - [Model (View 1)]
Meshed (Fluid) vs. Meshed (Deformable)
Meshed (Fluid) vs. Meshed (Rigid)
Meshed (Fluid) vs. Symmetry
Meshed (Deformable) vs. Meshed (Deformable)
Meshed (Deformable) vs. Meshed (Rigid)
Meshed (Deformable) vs. Symmetry
Meshed (Rigid) vs. Meshed (Rigid)
Meshed (Rigid) vs. Symmetry
1. Create contact interaction between plate and air block.
2. Set name
3. Click OK button.
Contact Interaction Properties
Name interact_plate_air
Type Meshed (Fluid)
Meshed (Deformable)
Reset
OK

Contact

1. Create Contact Table.

2. Activate interaction between bottom plate and air block.

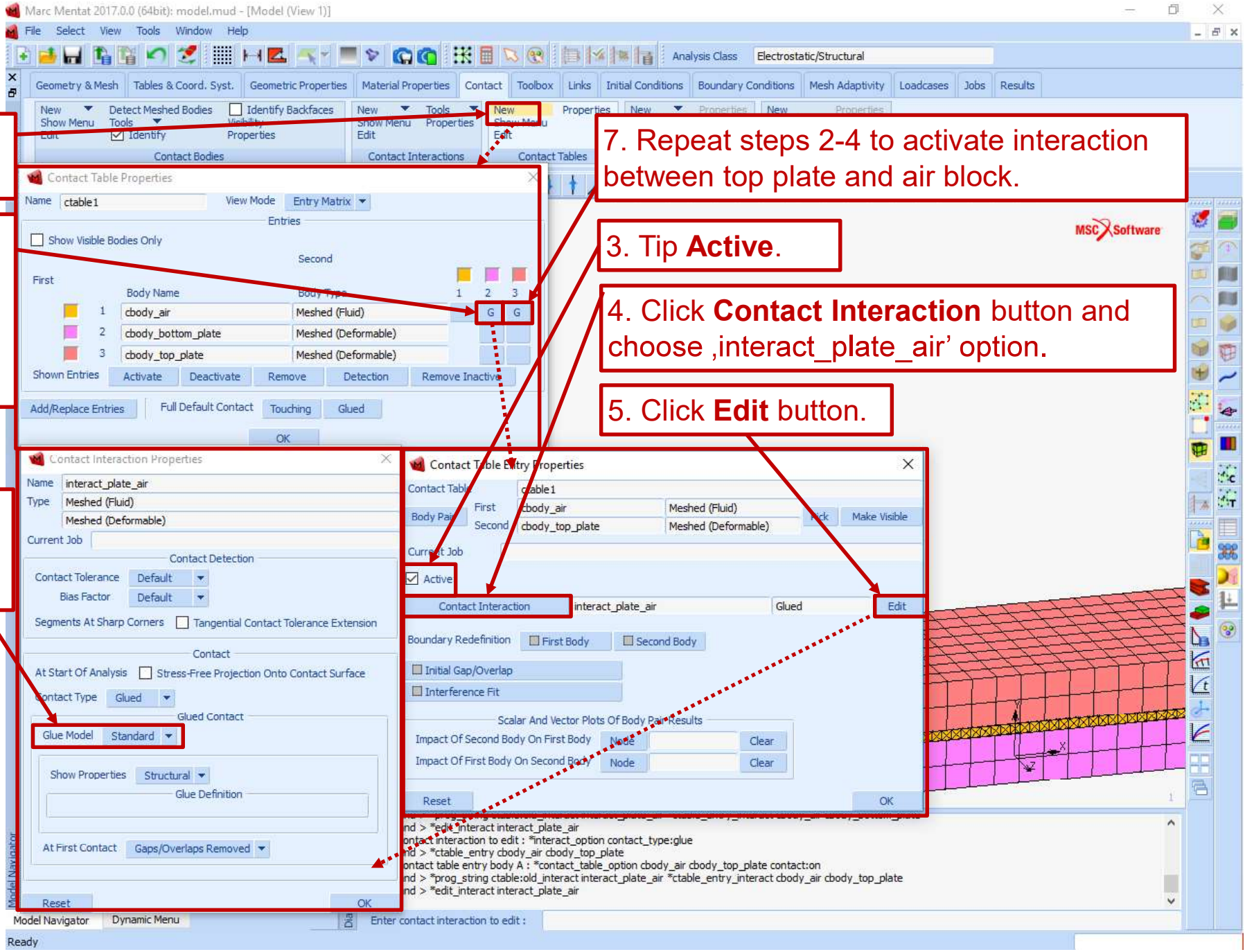
6. Change ‚Contact Type' to ‚Glued'.

7. Repeat steps 2-4 to activate interaction between top plate and air block.

3. Tip **Active**.

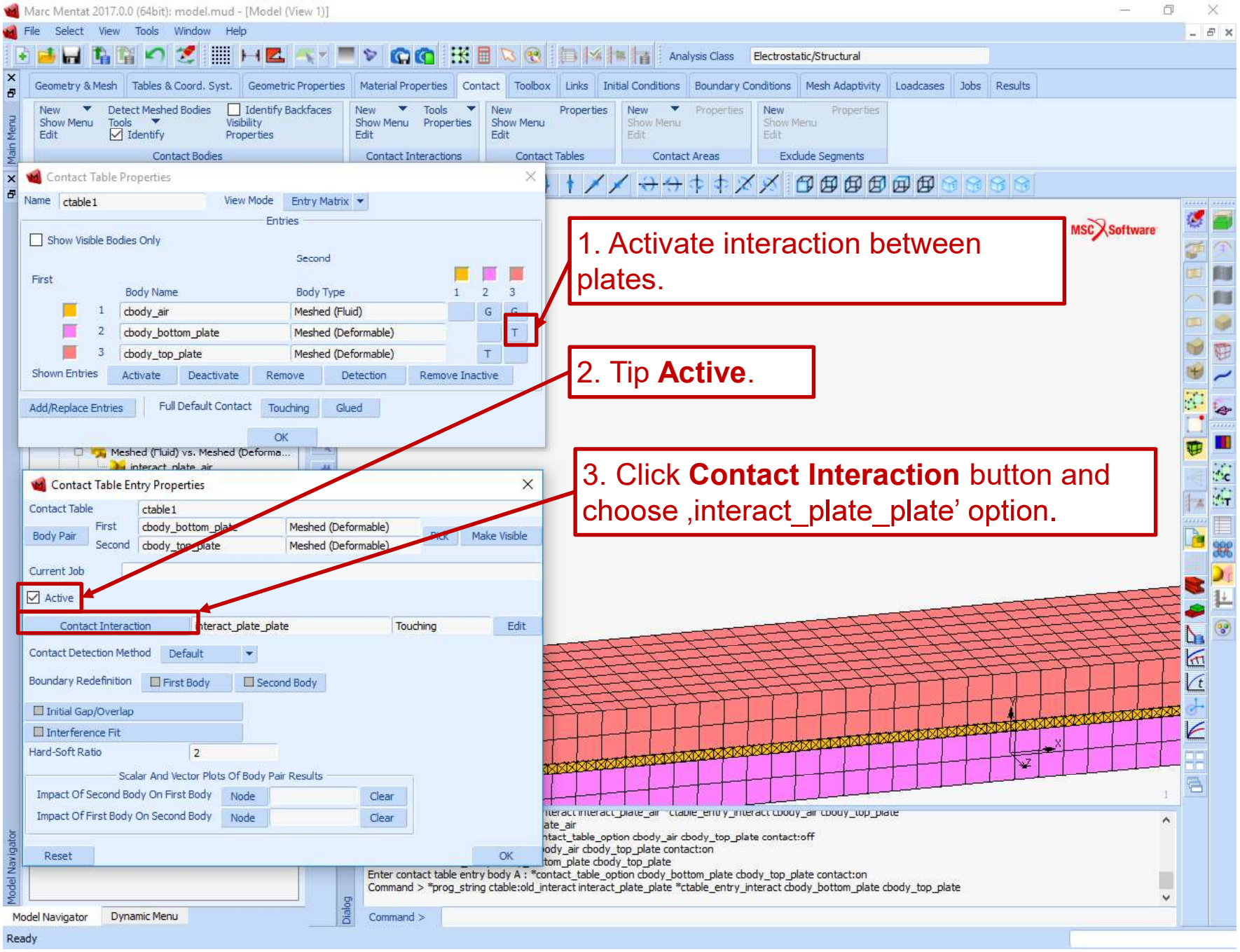
4. Click **Contact Interaction** button and choose ‚interact_plate_air' option.

5. Click **Edit** button.



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Contact


1. Activate interaction between plates.
2. Tip **Active**.
3. Click **Contact Interaction** button and choose ‚interact_plate_plate' option.
Contact Table Properties
Contact Table Entry Properties
Active
Contact Interaction
interact_plate_plate
Touching
Edit

Boundary conditions

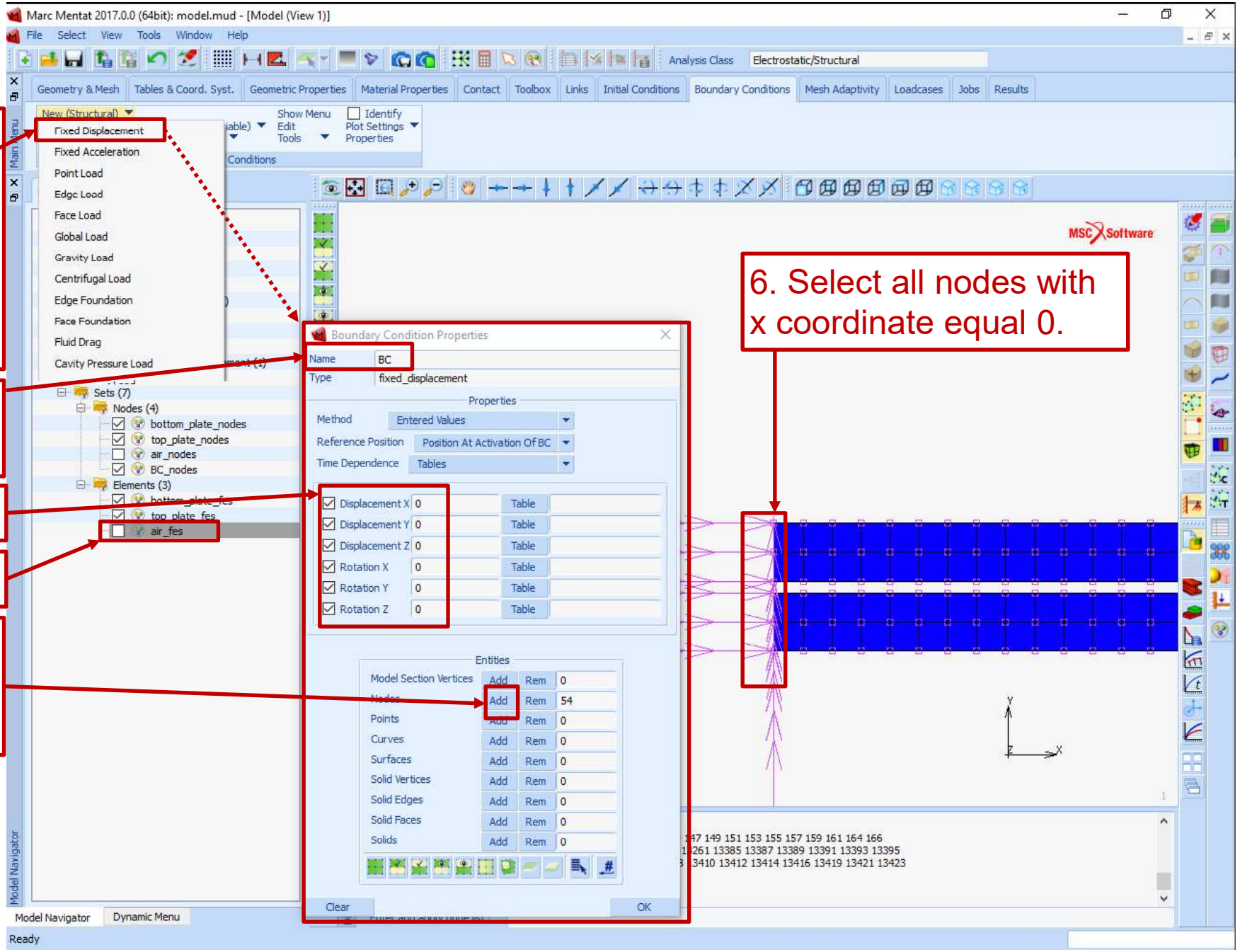
1. Define boundary conditio by choosing ‚Fixed Displacement' option.

2. Set name: BC.

3. Tick all DOF.

4. Untick air_fes

5. Add nodes by clicking **Add** button.



6. Select all nodes with x coordinate equal 0.

Boundary conditions

1. Define electric potantial by clicking Fixed Electric Potential option.

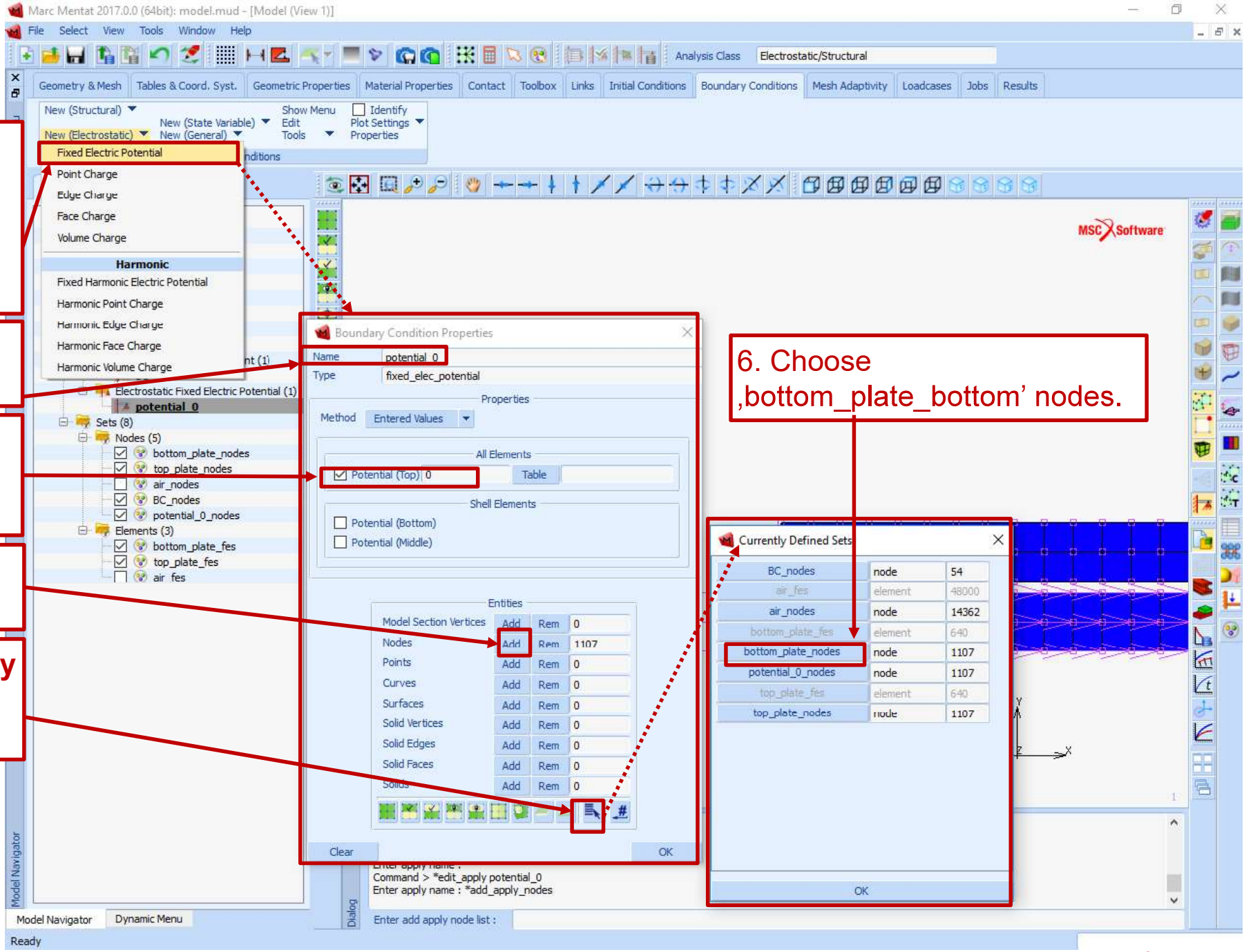
2. Set name: potantial_0.

3. Tick ‚Potential(Top)' option.

4. Click **Add** button.

5. Click **Currently defined set(s)** buttons.

6. Choose ‚bottom_plate_bottom' nodes.



Boundary conditions

1. Define electric potantial by clicking Fixed Electric Potential option.

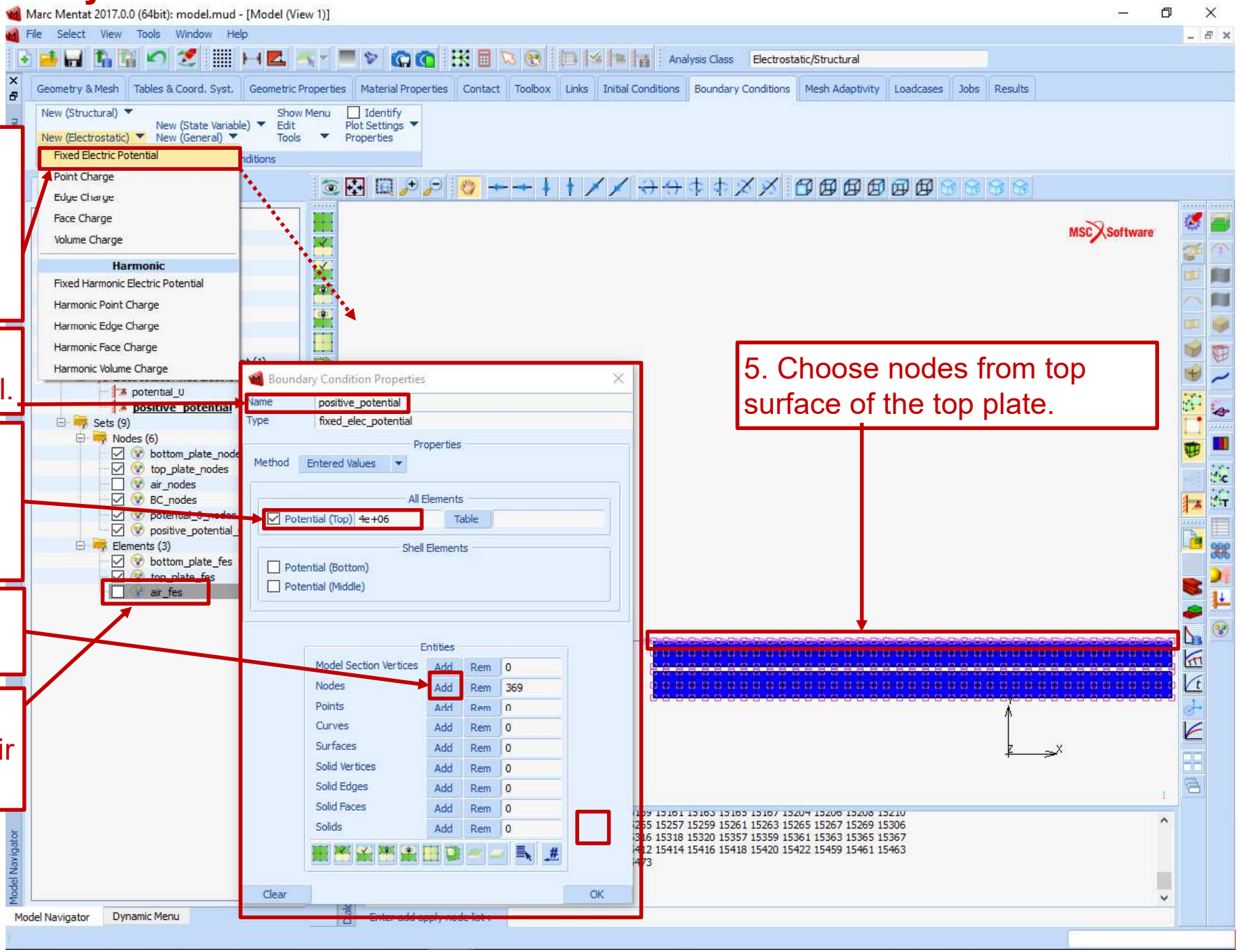
2. Set name: positive_potantial.

3. Tick ‚Potential(Top)' option and set 4e+6.

4. Click **Add** button.

6. Tick ‚air_fes' option to make air block visible.

5. Choose nodes from top surface of the top plate.



Loadcase

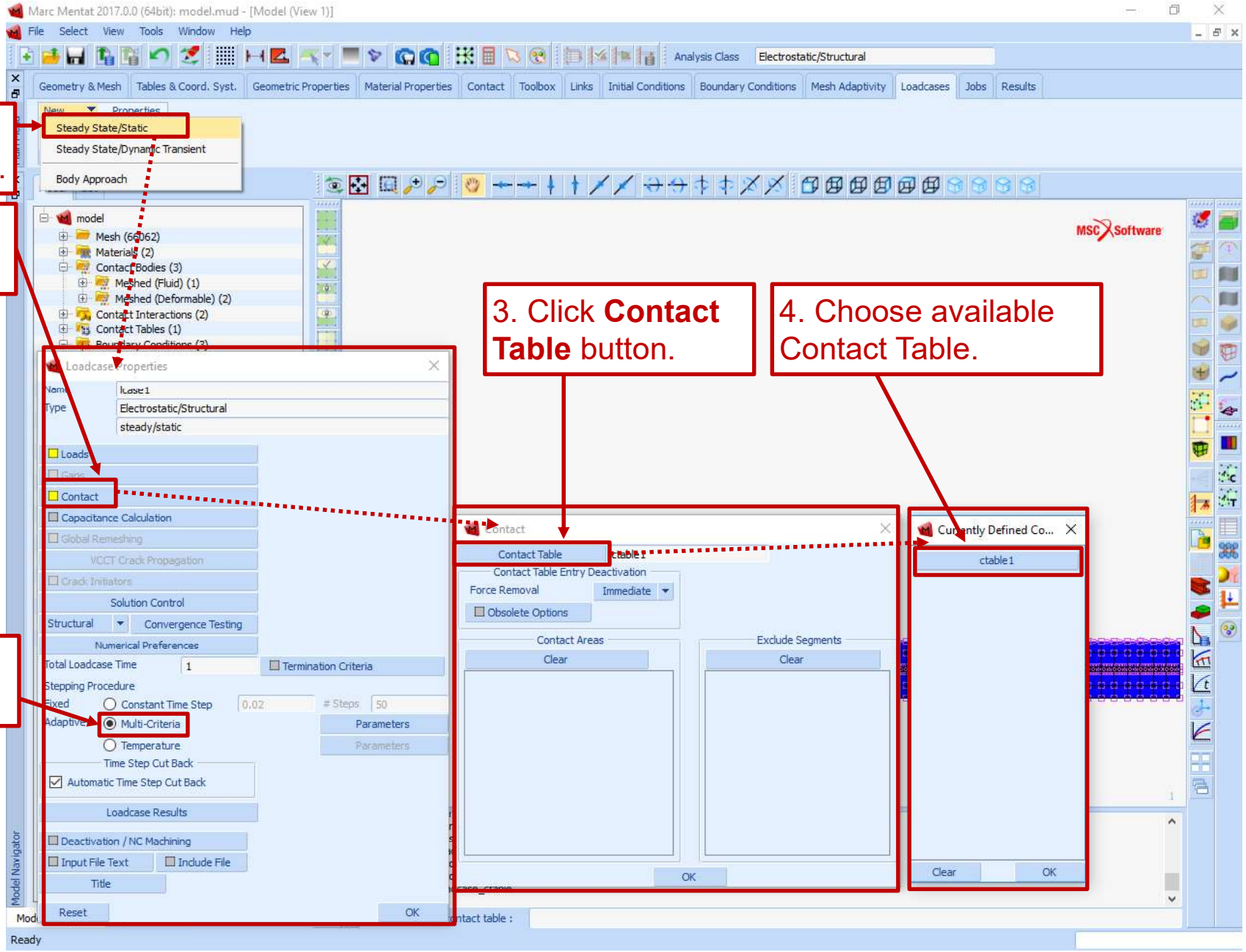
1. Create Steady State/Static Loadcase.

2. Click **Contact** button.

3. Click **Contact Table** button.

4. Choose available Contact Table.

5. Tick ‚Multi-Criteria' option.



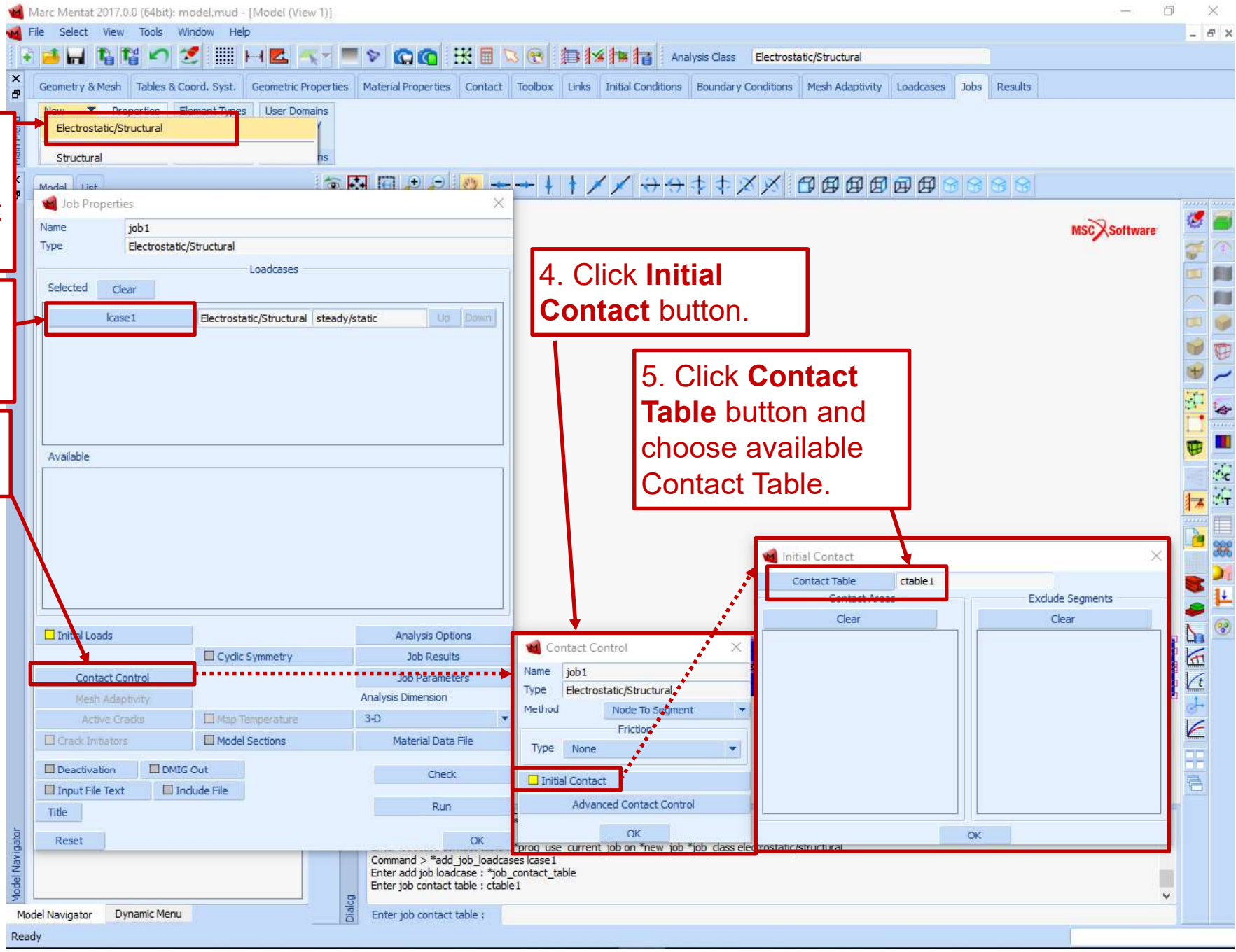
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Job

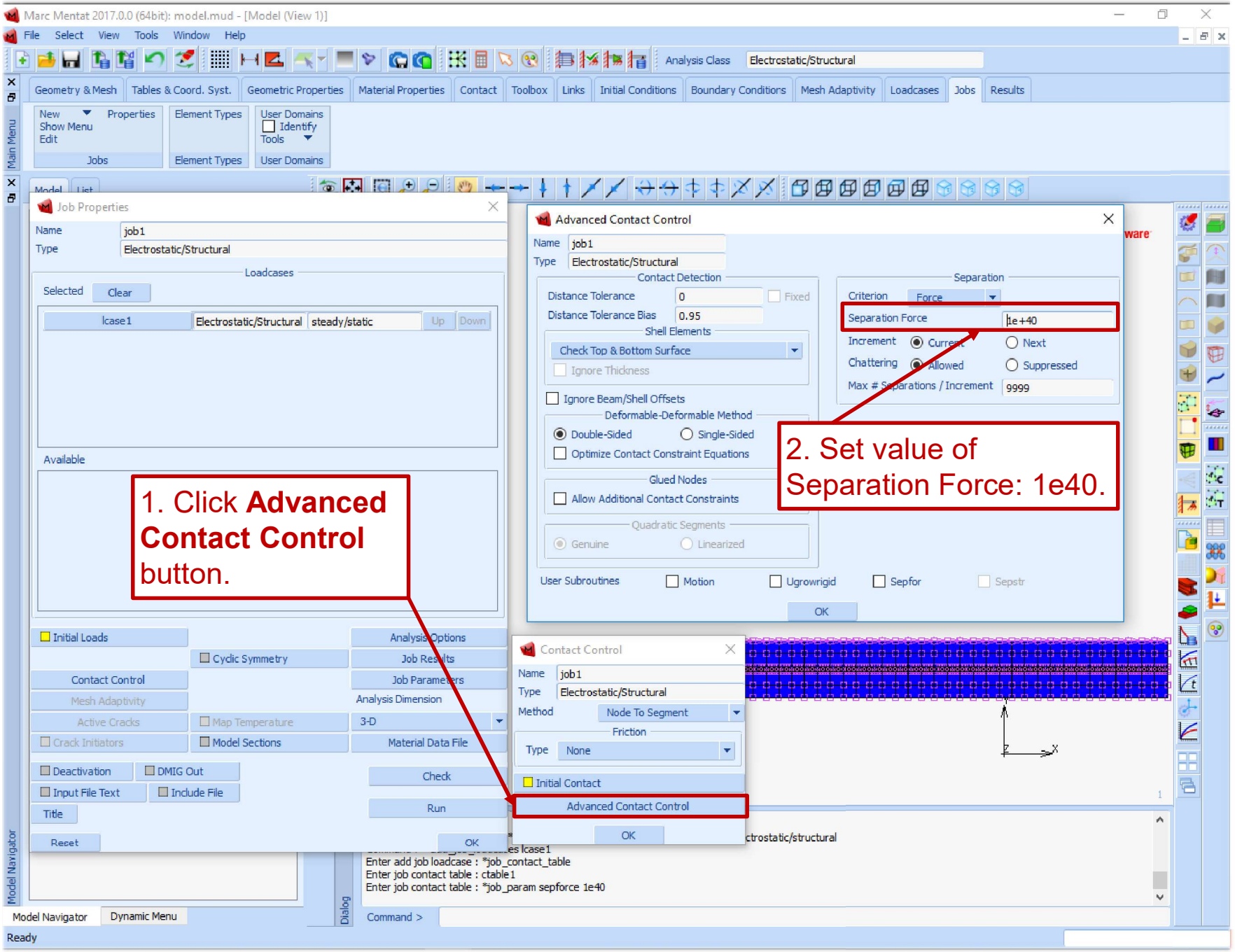
1. Define job by choosing ‚Electrostatic/Structural' option.

2. Move loadcase from ‚Available' to ‚Selected' area.

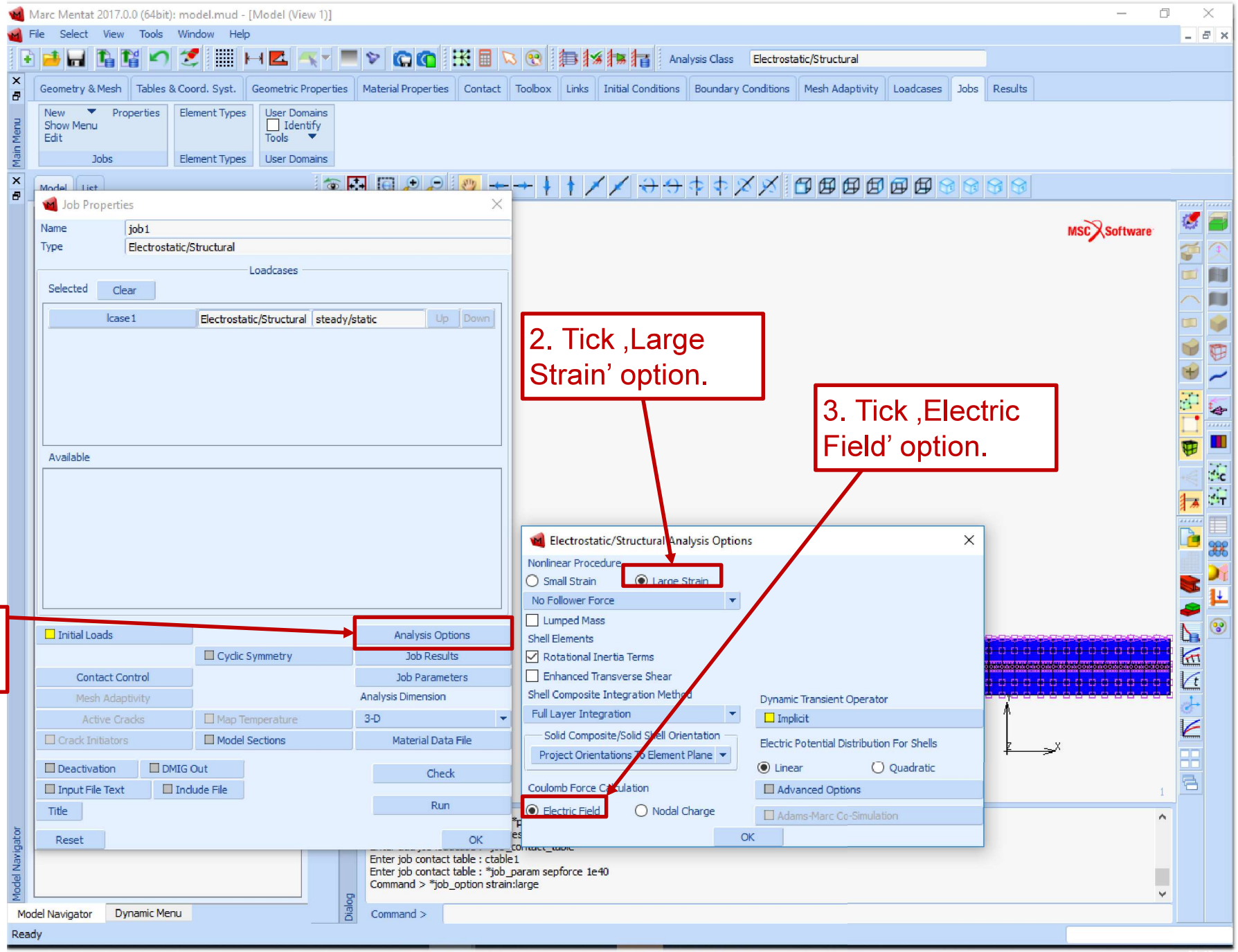
3. Click **Contact Control** button.



Job


1. Click **Advanced Contact Control** button.
2. Set value of Separation Force: 1e40.
Job Properties
Advanced Contact Control
Contact Control
Separation Force
1e+40
Advanced Contact Control
Enter add job loadcase : *job_contact_table
Enter job contact table : ctable1
Enter job contact table : *job_param sepforce 1e40

Job


Marc Mentat 2017.0.0 (64bit): model.mud - [Model (View 1)]
File Select View Tools Window Help
Analysis Class Electrostatic/Structural
Geometry & Mesh Tables & Coord. Syst. Geometric Properties Material Properties Contact Toolbox Links Initial Conditions Boundary Conditions Mesh Adaptivity Loadcases Jobs Results
Job Properties
Name job1
Type Electrostatic/Structural
Loadcases
Selected Clear
lcase1 Electrostatic/Structural steady/static Up Down
Available
Initial Loads
Analysis Options
Cyclic Symmetry
Job Results
Contact Control
Job Parameters
Mesh Adaptivity
Analysis Dimension
Active Cracks
Map Temperature
3-D
Crack Initiators
Model Sections
Material Data File
Deactivation
DMIG Out
Check
Input File Text
Include File
Run
Title
Reset
OK
Electrostatic/Structural Analysis Options
Nonlinear Procedure
Small Strain
Large Strain
No Follower Force
Lumped Mass
Shell Elements
Rotational Inertia Terms
Enhanced Transverse Shear
Shell Composite Integration Method
Full Layer Integration
Solid Composite/Solid Shell Orientation
Project Orientations To Element Plane
Dynamic Transient Operator
Implicit
Electric Potential Distribution For Shells
Linear
Quadratic
Coulomb Force Calculation
Electric Field
Nodal Charge
Advanced Options
Adams-Marc Co-Simulation
OK
Enter job contact table : ctable1
Enter job contact table : *job_param sepforce 1e40
Command > *job_option strain:large
Model Navigator
Dynamic Menu
Command >
Ready
2. Tick ‚Large Strain' option.
3. Tick ‚Electric Field' option.
1. Click **Analysis Option** button.

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RUN JOB AND ATTACH RESULTS TO REPORT